

Experiment Report: Istio and Bookinfo Deployment on Windows

1. Objective The purpose of this experiment is to deploy Istio service mesh on a Kubernetes cluster running on Windows, install the Bookinfo sample application, configure a gateway for external access, and monitor the services using Kiali dashboard. The experiment demonstrates Istio's core components, sidecar injection, Gateway API, and observability features.
2. Environment
 - Operating System: Windows 10/11
 - Kubernetes: Minikube / KIND / existing cluster
 - Istio version: 1.28.0
 - Tools: kubectl, PowerShell, Istioctl
3. Experiment Steps and Observations

Step 1: Install Istio [Insert Screenshot] Observation: Istio installation completed without errors.

Step 2: Enable automatic sidecar injection [Insert Screenshot] Observation: Namespace labeled successfully, sidecar injection enabled.

Step 3: Install Gateway API CRDs [Insert Screenshot] Observation: CRDs for Gateways and HTTPRoutes created successfully.

Step 4: Deploy Bookinfo application [Insert Screenshot] Observation: All services and pods created, pods READY 2/2.

Step 5: Deploy Bookinfo Gateway [Insert Screenshot] Observation: Gateway and HTTPRoute created, service type changed to ClusterIP.

Step 6: Verify Gateway [Insert Screenshot] Observation: Gateway exists and programmed successfully.

Step 7: Port-forward for external access [Insert Screenshot] Observation: Bookinfo accessible at localhost:8080, page loads correctly.

Step 8: Validate application inside cluster [Insert Screenshot] Observation: Productpage title retrieved, application running inside cluster.

Step 9: Install Kiali and other addons [Insert Screenshot] Observation: Kiali deployed successfully, dashboard accessible.

Step 10: View mesh topology in Kiali [Insert Screenshot] Observation: Traffic flows and service relationships visible.

Step 11: Send multiple requests to generate traces [Insert Screenshot] Observation: Requests succeeded, traces visible in Kiali and Jaeger.

Step 12: Confirm pods and services [Insert Screenshot] Observation: All pods READY 2/2, services running properly.

Step 13: Summary of results Observation: Istio core installed, Bookinfo deployed with sidecars, Gateway API configured, telemetry working.

4. Conclusion The experiment successfully deployed Istio on Windows with Bookinfo application, set up sidecar injection, Gateway API, and telemetry dashboards. The environment is now ready for further Istio feature evaluation.

```
PS C:\WINDOWS\system32> istioctl version
2025-11-18T10:53:46.553435Z    warn    default revision watcher not synced, falling back to "default"
Istio is not present in the cluster: no running Istio pods in namespace "istio-system"
2025-11-18T10:53:46.563728Z    warn    default revision watcher not synced, falling back to "default"
client version: 1.28.0
PS C:\WINDOWS\system32>
```

```
PS D:\istio\istio-1.28.0> istioctl install -f .\samples\bookinfo\demo-profile-no-gateways.yaml -y
>>
```



```
✓ Istio core installed ▲
✓ Istiod installed ☁
✓ Installation complete
```

```
PS D:\istio\istio-1.28.0> kubectl label namespace default istio-injection=enabled --overwrite
namespace/default labeled
```

```
PS D:\istio\istio-1.28.0> kubectl apply -f .\samples\bookinfo\platform\kube\bookinfo.yaml
service/details created
serviceaccount/bookinfo-details created
deployment.apps/details-v1 created
service/ratings created
serviceaccount/bookinfo-ratings created
deployment.apps/ratings-v1 created
service/reviews created
serviceaccount/bookinfo-reviews created
deployment.apps/reviews-v1 created
deployment.apps/reviews-v2 created
deployment.apps/reviews-v3 created
service/productpage created
serviceaccount/bookinfo-productpage created
deployment.apps/productpage-v1 created
PS D:\istio\istio-1.28.0>
```

```
PS D:\istio\istio-1.28.0> kubectl apply -f D:\istio\standard-install.yaml
Warning: unrecognized format "int32"
Warning: unrecognized format "int64"
customresourcedefinition.apiextensions.k8s.io/backendtlspolicies.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/gatewayclasses.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/gateways.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/grpcroutes.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/httproutes.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/referencegrants.gateway.networking.k8s.io created
PS D:\istio\istio-1.28.0>
```

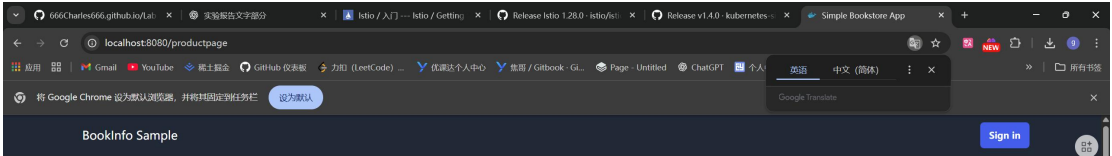
```
PS D:\istio\istio-1.28.0> kubectl get crd gateways.gateway.networking.k8s.io
NAME                                CREATED AT
gateways.gateway.networking.k8s.io  2025-11-18T11:10:55Z
PS D:\istio\istio-1.28.0> kubectl get crd httproutes.gateway.networking.k8s.io
NAME                                CREATED AT
httproutes.gateway.networking.k8s.io 2025-11-18T11:10:56Z
```

```
PS D:\istio\istio-1.28.0> kubectl apply -f D:\istio\istio-1.28.0\samples\bookinfo\gateway-api\bookinfo-gateway.yaml
gateway.gateway.networking.k8s.io/bookinfo-gateway created
httproute.gateway.networking.k8s.io/bookinfo created
PS D:\istio\istio-1.28.0> kubectl get gateway -n default
NAME          CLASS  ADDRESS  PROGRAMMED  AGE
bookinfo-gateway  istio  False    False       11s
PS D:\istio\istio-1.28.0>
```

```
PS D:\istio\istio-1.28.0> kubectl get crd gateways.gateway.networking.k8s.io
Error from server (NotFound): customresourcedefinitions.apiextensions.k8s.io "gateways.gateway.networking.k8s.io" not found
PS D:\istio\istio-1.28.0> kubectl get crd gateways.gateway.networking.k8s.io
Error from server (NotFound): customresourcedefinitions.apiextensions.k8s.io "gateways.gateway.networking.k8s.io" not found
PS D:\istio\istio-1.28.0> kubectl get crd gateways.gateway.networking.k8s.io
Error from server (NotFound): customresourcedefinitions.apiextensions.k8s.io "gateways.gateway.networking.k8s.io" not found
PS D:\istio\istio-1.28.0> kubectl kustomize "github.com/kubernetes-sigs/gateway-api/config/crd?ref=v1.4.0" | kubectl apply -f -
error: hit 2fs timeout running "D:\浏览器下载\下载\Git\cmd\git.exe fetch --depth=1 https://github.com/kubernetes-sigs/gateway-api v1.4.0"
error: no objects passed to apply
PS D:\istio\istio-1.28.0> ping github.com

正在 Ping github.com [20.205.243.166] 具有 32 字节的数据:
来自 20.205.243.166 的回复: 字节=32 时间=142ms TTL=106
来自 20.205.243.166 的回复: 字节=32 时间=108ms TTL=106
来自 20.205.243.166 的回复: 字节=32 时间=99ms TTL=106
来自 20.205.243.166 的回复: 字节=32 时间=99ms TTL=106

20.205.243.166 的 Ping 统计信息:
    数据包: 已发送 = 4, 已接收 = 4, 丢失 = 0 (0% 丢失),
    往返行程的估计时间 (以毫秒为单位):
        最短 = 90ms, 最长 = 142ms, 平均 = 110ms
PS D:\istio\istio-1.28.0> kubectl kustomize "github.com/kubernetes-sigs/gateway-api/config/crd?ref=v1.4.0" | kubectl apply -f -
error: failed to run "D:\浏览器下载\下载\Git\cmd\git.exe fetch --depth=1 https://github.com/kubernetes-sigs/gateway-api v1.4.0": fatal: unable to access 'https://github.com/kubernetes-sigs/gateway-api/': Error 1
in the HTTP2 framing layer
PS D:\istio\istio-1.28.0> exit status 128
error: no objects passed to apply
PS D:\istio\istio-1.28.0> kubectl apply -f D:\istio\gateway-crds-standard.yaml
PS D:\istio\istio-1.28.0> kubectl apply -f D:\istio\gateway-crds-standard.yaml
error: the path "D:\istio\gateway-crds-standard.yaml" does not exist
error: the path "D:\istio\gateway-crds-standard.yaml" does not exist
PS D:\istio\istio-1.28.0> kubectl apply -f D:\istio\crds-standard.yaml
error: the path "D:\istio\crds-standard.yaml" does not exist
PS D:\istio\istio-1.28.0> kubectl apply -f D:\istio\standard-install.yaml
Warning: unrecognized format "int64"
Warning: unrecognized format "int64"
customresourcedefinition.apiextensions.k8s.io/backendtlspolicies.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/gatewayclasses.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/gateways.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/grproutes.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/httproutes.gateway.networking.k8s.io created
customresourcedefinition.apiextensions.k8s.io/referencegrants.gateway.networking.k8s.io created
PS D:\istio\istio-1.28.0> kubectl get crd gateways.gateway.networking.k8s.io
NAME          CREATED AT
gateways.gateway.networking.k8s.io 2025-11-18T11:10:55Z
PS D:\istio\istio-1.28.0> kubectl get crd httproutes.gateway.networking.k8s.io
NAME          CREATED AT
httproutes.gateway.networking.k8s.io 2025-11-18T11:10:56Z
PS D:\istio\istio-1.28.0> kubectl apply -f D:\istio\istio-1.28.0\samples\bookinfo\gateway-api\bookinfo-gateway.yaml
gateway.gateway.networking.k8s.io/bookinfo-gateway created
httproute.gateway.networking.k8s.io/bookinfo created
PS D:\istio\istio-1.28.0> kubectl get gateway - default
NAME          CLASS          ADDRESS          PROGRAMMED  AGE
bookinfo-gateway  istio          False           11s
PS D:\istio\istio-1.28.0> kubectl port-forward svc/bookinfo-gateway-istio 8080:80 -n default
Forwarding from 127.0.0.1:8080 -> 80
Handling connection for 8080
Handling connection for 8080
```



The Comedy of Errors

Wikipedia Summary: The Comedy of Errors is one of **William Shakespeare's** early plays. It is his shortest and one of his most farcical comedies, with a major part of the humour coming from slapstick and mistaken identity, in addition to puns and word play.

[Learn more about Istio ->](#)

Book Details

ISBN-10	Publisher	Pages	Type	Language
1234567890	PublisherA	200	paperback	English

Book Reviews

"An extremely entertaining play by Shakespeare. The slapstick humour is refreshing!"

 Reviewer1

Reviews served by: reviews-v1-8cf7b9cc5-82737

"Absolutely fun and entertaining. The play lacks thematic depth when compared to other plays by Shakespeare."

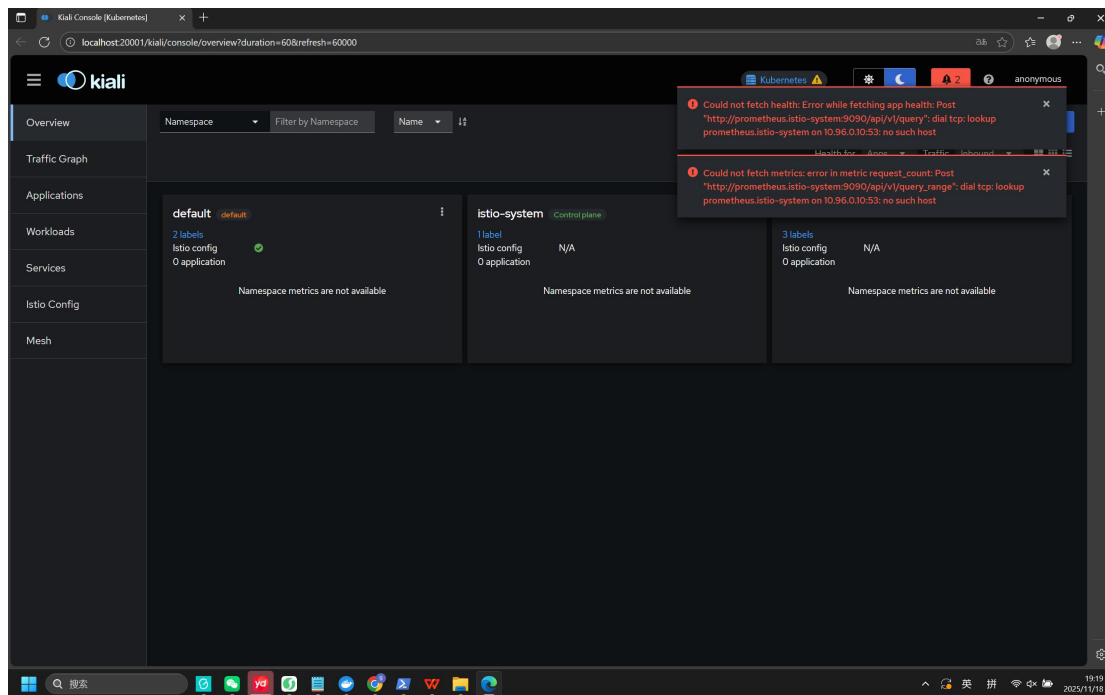
 Reviewer2

Reviews served by: reviews-v1-8cf7b9cc5-82737

```
PS C:\WINDOWS\system32> $pod = kubectl get pod -l app=ratings -o jsonpath='{.items[0].metadata.name}'
PS C:\WINDOWS\system32> kubectl exec $pod -c ratings -- curl -sS productpage:9080/productpage | Select-String -Pattern '<title>.*</title>'

<title>Simple Bookstore App</title>

PS C:\WINDOWS\system32> kubectl apply -f .\samples\addons\kiali.yaml
error: the path ".\samples\addons\kiali.yaml" does not exist
PS C:\WINDOWS\system32> kubectl rollout status deployment/kiali -n istio-system
Error from server (NotFound): deployments.apps "kiali" not found
PS C:\WINDOWS\system32> cd D:\istio\istio-1.28.0
PS D:\istio\istio-1.28.0> kubectl apply -f .\samples\addons\kiali.yaml
serviceaccount/kiali created
configmap/kiali created
clusterrole.rbac.authorization.k8s.io/kiali created
clusterrolebinding.rbac.authorization.k8s.io/kiali created
service/kiali created
deployment.apps/kiali created
PS D:\istio\istio-1.28.0> kubectl rollout status deployment/kiali -n istio-system
Waiting for deployment "kiali" rollout to finish: 0 out of 1 new replicas have been updated...
Waiting for deployment "kiali" rollout to finish: 0 of 1 updated replicas are available...
deployment "kiali" successfully rolled out
PS D:\istio\istio-1.28.0>
```



```
PS D:\istio\istio-1.28.0> Invoke-WebRequest -Uri "http://localhost:8080/productpage" -UseBasicParsing
```

```

StatusCode      : 200
StatusDescription : OK
Content         :
    <meta charset="utf-8">
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Simple Bookstore App</title>

    <script src=...
RawContent      : HTTP/1.1 200 OK
                  vary: Cookie
                  x-envoy-upstream-service-time: 152
                  Content-Length: 15066
                  Content-Type: text/html; charset=utf-8
                  Date: Tue, 18 Nov 2025 11:25:43 GMT
                  Server: istio-envoy

    <meta cha...
Forms           :
Headers         : {[vary, Cookie], [x-envoy-upstream-service-time, 152], [Content-Length, 15066], [Content-Type, text/html; charset=utf-8]...}
Images          : {}
InputFields     : {}
Links           : @{{outerHTML=<a href="#" class="text-white px-3 py-2 text-lg font-medium" aria-current="page">BookI
                  nfo Sample</a>; tagName=A; href=#; class=text-white px-3 py-2 text-lg font-medium; aria-current=pag
                  e}, @{{outerHTML=<a href="https://istio.io" target="_blank" class="font-semibold leading-6 text-blue
                  -600 hover:text-blue-500">Start here</a>; tagName=A; href=https://istio.io; target=_blank; class=fo
                  nt-semibold leading-6 text-blue-600 hover:text-blue-500}, @{{outerHTML=<a href="https://en.wikipedia
                  .org/wiki/The_Comedy_of_Errors">Wikipedia Summary</a>; tagName=A; href=https://en.wikipedia.org/wik
                  i/The_Comedy_of_Errors}, @{{outerHTML=<a href="https://istio.io" target="_blank" class="text-sm font
                  -semibold leading-6 text-blue-600 hover:text-blue-700">Learn more about Istio <span aria-hidden="tr
                  ue">+</span></a>; tagName=A; href=https://istio.io; target=_blank; class=text-sm font-semibold lea
                  ding-6 text-blue-600 hover:text-blue-700}}
ParsedHtml      :
RawContentLength : 15066

```

```

PS D:\istio\istio-1.28.0> istioctl uninstall --purge -y
All Istio resources will be pruned from the cluster.

Removed apps/v1, Kind=Deployment/istiod.istio-system.
Removed /v1, Kind=Service/istiod.istio-system.
Removed /v1, Kind=Service/istiod-revision-tag-default.istio-system.
Removed /v1, Kind=ConfigMap/istio.istio-system.
Removed /v1, Kind=ConfigMap/istio-sidecar-injector.istio-system.
Removed /v1, Kind=ConfigMap/values.istio-system.
Removed /v1, Kind=Pod/istiod-d468976dc-88chd.istio-system.
Removed /v1, Kind=ServiceAccount/istio-reader-service-account.istio-system.
Removed /v1, Kind=ServiceAccount/istiod.istio-system.
Removed rbac.authorization.k8s.io/v1, Kind=RoleBinding/istiod.istio-system.
Removed rbac.authorization.k8s.io/v1, Kind=Role/istiod.istio-system.
Removed admissionregistration.k8s.io/v1, Kind=MutatingWebhookConfiguration/istio-revision-tag-default..
Removed admissionregistration.k8s.io/v1, Kind=ValidatingWebhookConfiguration/istio-sidecar-injector..
Removed admissionregistration.k8s.io/v1, Kind=ValidatingWebhookConfiguration/istio-validator-istio-system..
Removed rbac.authorization.k8s.io/v1, Kind=ClusterRole/istio-reader-clusterrole-istio-system..
Removed rbac.authorization.k8s.io/v1, Kind=ClusterRole/istiod-clusterrole-istio-system..
Removed rbac.authorization.k8s.io/v1, Kind=ClusterRole/istiod-gateway-controller-istio-system..
Removed rbac.authorization.k8s.io/v1, Kind=ClusterRoleBinding/istio-reader-clusterrole-istio-system..
Removed rbac.authorization.k8s.io/v1, Kind=ClusterRoleBinding/istiod-clusterrole-istio-system..
Removed rbac.authorization.k8s.io/v1, Kind=ClusterRoleBinding/istiod-gateway-controller-istio-system..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/authorizationpolicies.security.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/destinationrules.networking.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/envoyfilters.networking.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/gateways.networking.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/peerauthentications.security.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/proxyconfigs.networking.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/requestauthentications.security.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/serviceentries.networking.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/sidecars.networking.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/telemetries.telemetry.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/virtualservices.networking.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/wasmplugins.extensions.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/workloadentries.networking.istio.io..
Removed apiextensions.k8s.io/v1, Kind=CustomResourceDefinition/workloadgroups.networking.istio.io..
✓ Uninstall complete
PS D:\istio\istio-1.28.0> kubectl delete namespace istio-system
namespace "istio-system" deleted
PS D:\istio\istio-1.28.0> kubectl delete -f .\samples\bookinfo\platform\kube\bookinfo.yaml
service "details" deleted
serviceaccount "bookinfo-details" deleted
deployment.apps "details-v1" deleted
service "ratings" deleted
serviceaccount "bookinfo-ratings" deleted
deployment.apps "ratings-v1" deleted
service "reviews" deleted
serviceaccount "bookinfo-reviews" deleted
deployment.apps "reviews-v1" deleted
deployment.apps "reviews-v2" deleted
deployment.apps "reviews-v3" deleted
service "productpage" deleted
serviceaccount "bookinfo-productpage" deleted
deployment.apps "productpage-v1" deleted
PS D:\istio\istio-1.28.0>

```